

Q Show Bottom Up Parsing of the string "id + id * id" using the following grammar. (7 Marks)

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow E \mid \text{id}$$

⇒ What is Bottom-up Parsing?

- Bottom-up parser starts from the input string (terminals) and reduces it to the start symbol.

Two actions for Bottom-up parsing:

(1) shift → move input to stack

(2) Reduce → replace symbols using grammar rules.

⇒ Input : id + id * id \$
\$ means end of input.

① shift first id

∴ stack : id

input : + id * id \$

Tree : id

→ as $F \rightarrow \text{id}$, reduce id to F

∴ $\text{id} \rightarrow F$

now, because $T \rightarrow F$,

reduce $F \rightarrow T$

∴ stack : T

now, because $E \rightarrow T$

reduce $T \rightarrow E$

∴ stack : E

② shift +

∴ stack : E +

③ shift id

∴ stack : E + id

Parse Tree

id

F
|
id

T
|
F
|
id

E
|
T
|
F
|
id

→ as $F \rightarrow id$, reduce id to F

$\therefore id \rightarrow F \rightarrow E + F$

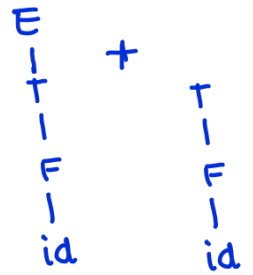
now, because $T \rightarrow F$,

reduce $F \rightarrow T$

$\therefore \text{stack} : E + T$

→ now, we do NOT reduce to E
because $*$ has higher priority.

Parse Tree



④ shift $*$

$\therefore \text{stack} : E + T *$

⑤ shift third id

$\therefore \text{stack} : E + T * id$

→ as $F \rightarrow id$, reduce id to F

$\therefore id \rightarrow F$

$\therefore \text{stack} : E + T * F$

→ reduce $T * F \rightarrow T$

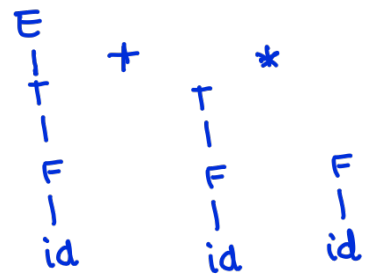
$\text{stack} : E + T$

→ as $E \rightarrow E + T$

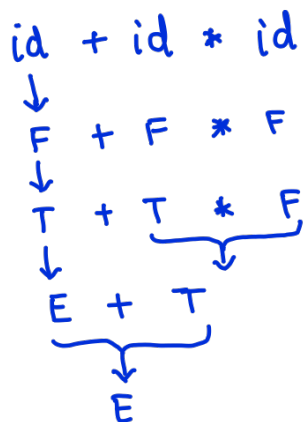
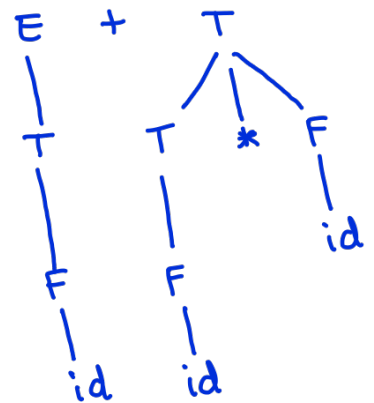
$\therefore \text{stack} : E$

→ as only E remains on
stack, string is accepted.

Parse Tree



Parse Tree



Parse Tree

